Phase 12C Muse S Athena Capture Pack

V7/V8-style visual: state, control, drift variable, alignment

FULL CAPTURES	WINDOWING	ATHENA PACKETS	DECODED ROWS	MAPPED LANES
5	**60 / 120 / 60**	**32,347**	**3.25M**	**5**
3 mirror + 2 controls	baseline / condition / post	sensor-bus packets across pack	engineering sample rows	EEG, optical, motion, quality, status

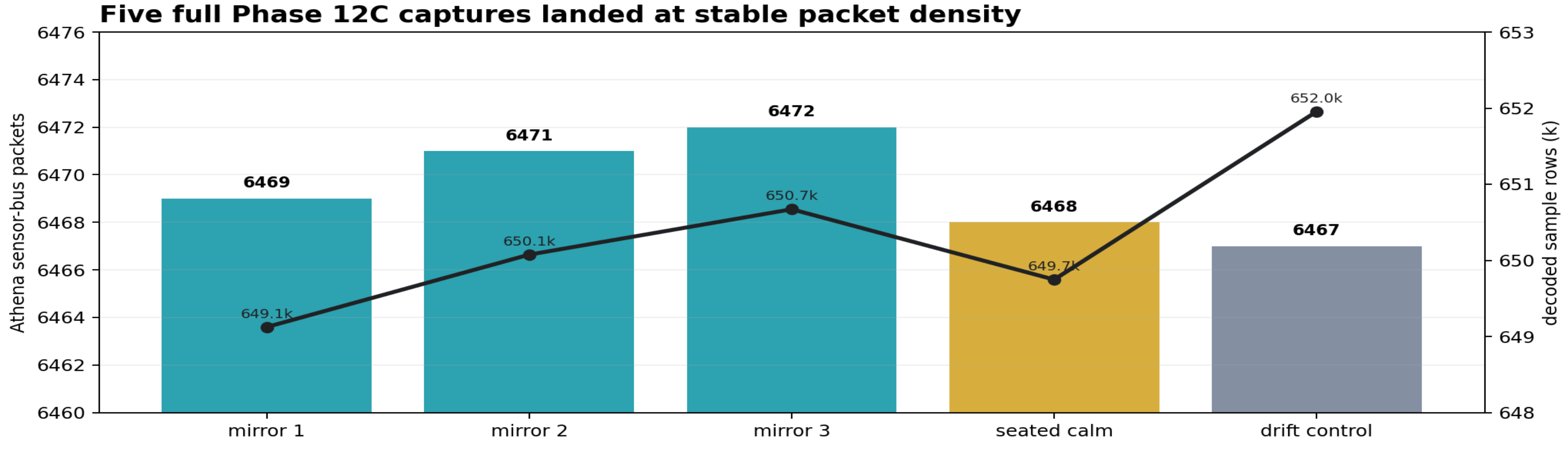
State / Control / Drift / Alignment Read

STATE	mirror_coherence condition state
CONTROL	seated_calm + drift_control comparators
DRIFT VARIABLE	IMU motion + DRL/reference quality travel with the signal
ALIGNMENT	condition-minus-baseline mirror average vs control average

Visual read: the administered state is mapped against controls while drift lanes remain attached to the alignment read.

Five-Run Matrix

Run	State / control	Packets	Samples	Alignment read
mirror 1	mirror_coherence	6469	649.1k	**strong shift + artifact lane**
mirror 2	mirror_coherence	6471	650.1k	**cleaner repeat**
mirror 3	mirror_coherence	6472	650.7k	**cleaner repeat**
seated calm	seated_calm	6468	649.7k	**reference lane**
drift control	drift_control	6467	652.0k	**drift comparator**



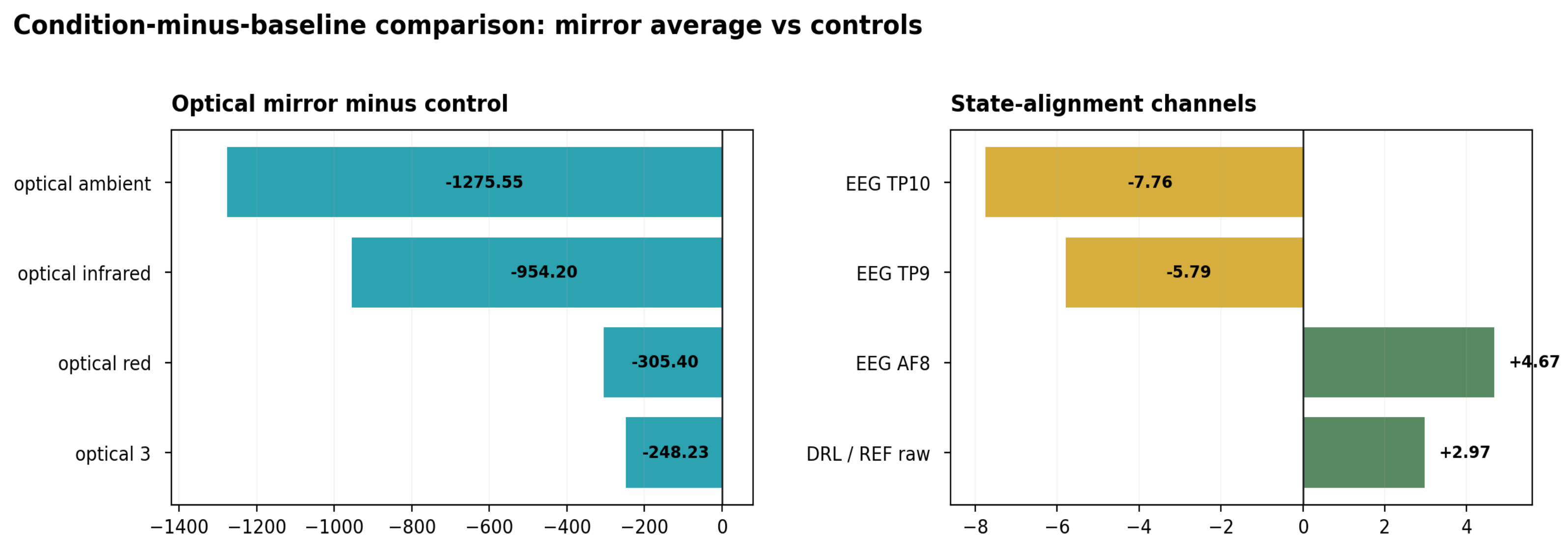
Capture-To-Readout Spine

Muse S Athena — native adapter — stream-ready — raw archive

Athena decoder — lane table — 12C windows — B.A.S.I.S. bus

Phase 12C Alignment Readout

Mirror average versus controls, with drift variable carried inside the evidence layer



Locked Read

- Five full Muse S Athena sessions reached the same packet-density scale.
- Decoded engineering lanes recur across baseline / condition / post windows.
- Mirror average separates from controls in optical, quality, and selected EEG channels.
- The drift variable is explicit: gyro z, IMU motion, and DRL/REF quality travel with the state lanes.

Condition Board

3 runs — **mirror_coherence**
administered state

1 run — **seated_calm**
calm control

1 run — **drift_control**
drift comparator

Window discipline: 60s baseline -> 120s condition -> 60s post.

B.A.S.I.S. Hub Pattern

Device adapter

Raw local archive

Decoder / normalizer

Lane table

Session manifest

State-vector bus

Reusable route for HRV, ECG, PPG / SpO2, GSR / EDA, CSV, API, OSC, LSL, webhook, and serial inputs.

Release Posture

Now — public evidence logs, counts, maps, visual packs

Private — capture code, raw packets, local device identifiers

After patent — review selected code for public release

Next evidence — motion/contact filtering + synchronized HRV

Evidence first. Public logging now. Raw implementation protected until patent completion.